Ihsan Ul Haq
Swat, KPK, Pakistan
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Website| Google Scholar | LinkedIn | GitHub

Research Interests

Renewable Energy Integration • Microgrids • Smart Grids • Intelligent Autonomous Systems • Computer Vision • Deep Learning • EV-grid Integration

Education

MSc Energy Engineering (Distinction)
University of Hull, United Kingdom | 2021–2022
Percentage: 72/100
Dissertation: *A Comparative Analysis of Deep Learning Models for Short-Term Load Forecasting.*
BSc Electrical Engineering
University of Engineering and Technology (UET), Peshawar, Pakistan | 2011–2015
GPA: 3.47/4.00
Final Year Project: *Design and Analysis of High Gain Microstrip Patch Antenna Using CST.*

Professional Experience

Research Associate (In Collaboration, Associated with UET Peshawar)
AI applications in electrical engineering and power systems, | 2023– Present
Primary School Teacher
Government Primary School (GPS) Malakana, Swat | 2020–2021
Subject Specialist (Mathematics)
Garland Model School and College | 2016–2020
Research Intern
PTCL Pakistan | Jul–Aug 2014

Selected Publications

Full Publication List →

Journal Articles

Khan, T., Mansoor, M., Iltaf, M., Alam, S., Farooq, M., **Ul Haq, I.**, & Ullah, K. (2026). Machine Learning-Based Renewable Energy Forecasting and Priority Load Management for Smart Energy Systems. *International Journal of Innovations in Science & Technology*, *8*(2), 786–802. https://doi.org/10.33411/IJIST/1881

Khan, A.R., **Haq, I.U.**, Farooq, M., Rehman, B.U., Ullah, K., Amir, M., Khan, M.K. and Khan, M.I. (2025) ‘Sustainable Energy Management in Smart Grids Using Machine Learning’, *Spectrum of Engineering Sciences*, 3(7), pp. 1332–1342. https://doi.org/10.5281/zenodo.16671247

Qasim, M., Akbarzai, J., Khan, A. R., Majeed, M. E., Farooq, M., **Haq, I. U.**, Rehman, B. U., Ullah, K., & Khan, M. K., “Machine Learning-Based Fault Detection in Three Phase Transmission Lines and Electric Machines”, Zenodo, Jul. 2024. doi: 10.5281/zenodo.16146039.

Conference Paper

Haq, I.U., Khalid, M. and Manzoor, U. (2023) ‘A Comparative Analysis of Deep Learning Models for Short-Term Load Forecasting’, *IEEE CASE 2023*, Auckland, New Zealand, pp. 1–7. doi:10.1109/CASE56687.2023.10260626.

Technical Skills

Programming: Python, MATLAB, C++
Machine Learning: TensorFlow, PyTorch, Keras, Scikit-learn, Deep Learning, Computer Vision
Data Science: Data Analysis, Data Visualization
Engineering Tools: CST Studio Suite, EndNote, Microsoft Visio, PowerPoint, PSpice

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Selected Certifications

- Renewable Energy Technology Fundamentals — University of Colorado Boulder
- Renewable Power & Electricity Systems — University of Colorado Boulder
- Advanced Learning Algorithms — DeepLearning.AI
- Reinforcement Learning — University of Alberta / DeepLearning.AI
- Mathematics for Machine Learning: Linear Algebra — Imperial College London
- An Intuitive Introduction to Probability — University of Zurich
- Python Data Structures — University of Michigan
- Convolutional Neural Networks — DeepLearning.AI

Awards & Memberships

- NICT Merit Scholarship, UET Peshawar (2011–2015)
- Pakistan Engineering Council (PEC), Membership No. 48533
- Saudi Council of Engineers, Membership No. 305735
- Academic Excellence Award (2010)

Languages

Pashto (Native) | English (Advanced) | Urdu (Advanced)

References

References are available upon request